



# Facile synthesis of iridium nanoparticles with superior peroxidase-like activity for colorimetric determination of $\text{H}_2\text{O}_2$ and xanthine



Malin Cui, Jingdan Zhou, Yuan Zhao, Qijun Song\*

The Key Laboratory of Food Colloids and Biotechnology, Ministry of Education, School of Chemical and Material Engineering, Jiangnan University, Wuxi 214122, China

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## ABSTRACT

A facile and efficient method was developed for the preparation of iridium nanoparticles (Ir NPs) with superior peroxidase-like activity. Under the catalytic action of the Ir NPs, the peroxidase substrate 3,3',5,5'-tetramethylbenzidine (TMB) can be oxidized by  $\text{H}_2\text{O}_2$  to form a blue-colored production oxTMB. Kinetic analysis indicated that the catalytic behavior was in accordance with the typical Michaelis-Menten kinetics and the catalytic reaction followed the ping-pong mechanism. Electron Spin Resonance (ESR) experiment showed that no hydroxyl radical formed in the reaction process. Peroxide substrates with different electron withdrawing ability (benzoyl peroxide,  $\text{H}_2\text{O}_2$  and artemisinin) were used to study the catalytic reactions and results indicated that  $\text{H}_2\text{O}_2$  can oxidize TMB with much faster rate than artemisinin and slower rate than benzoyl peroxide. These results suggest an electron transfer mechanism was involved in the system of TMB- $\text{H}_2\text{O}_2$ -Ir NPs. By using TMB as the colorimetric substrate,  $\text{H}_2\text{O}_2$  can be rapidly determined and the method was extended for the determination of xanthine based on its production of  $\text{H}_2\text{O}_2$  in the presence of xanthine oxidase. The prepared Ir NPs exhibit good stability in wide range of pH and temperature. The Ir NPs retained at least 90% of their initial catalytic activity after stored at ambient temperature for three months. The high specificity for  $\text{H}_2\text{O}_2$  and the excellent stability of the peroxidase-like Ir NPs showed the great application potential in biotechnology field.

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## 1. Introduction

Peroxidase mimics, including carbon nanomaterials [1–3], metallic oxides [4,5], metal chalcogenides [6] and metal nanomaterials [7], have attracted a great deal of research interest due to their tunable catalytic activity, excellent stability and biocompatibility. The intrinsic drawbacks of peroxidase, such as time-consuming and expensive production, sophisticated purification, and low stability to temperature variations can be largely avoided by using the nanomaterial based peroxidase mimics [8,9]. Peroxidase mimics of noble metal nanomaterials show potential application value in chemical analysis and biotechnology field. Such as Au [10], Pt [11] and Pd [12] nanomaterials were exploited as peroxidase-like labels for colorimetric determination of glucose. Mouse interleukin 2 (IL-2) was determined based on the peroxidase-like activity of Au@Pt nanomaterials through enzyme linked immunosorbent assay (ELISA) [13]. Colorimetric assay of human chronic myelogenous leukemia cell lines (K-562) was established using the Au@Pd nanoparticles

with peroxidase-like activity as labels [14]. Polyvinylpyrrolidone-capped Ir NPs (PVP-Ir NPs) with peroxidase-like activity were prepared for the first time in 2015, which can protect cells from  $\text{H}_2\text{O}_2$ -induced oxidative damage [15]. The catalytic mechanism of the PVP-Ir NPs was suggested to be an electron transfer process. However, little experimental evidences were provided to highlight the pathways. Furthermore, their method for the Ir NPs preparation was time consuming and the use of polymer PVP as the stabilizer may also hinder the catalytic activity of the prepared nanoparticles [16].

Hence in present study, a small molecule tannic acid was used as stabilizer to prepare the Ir NPs. Tannic acid is one kind of typical green amphiphilic molecule obtained from plants. The multiple O–H sites of tannic acid could weakly stabilized Ir NPs, which is important for mimics to exert the catalytic performance [17]. Moreover, the good water-solubility is beneficial for peroxidase mimics used in aqueous media. The Ir NPs stabilized with tannic acid can rapidly catalyze the oxidation of TMB to form blue oxTMB in the presence of  $\text{H}_2\text{O}_2$ . Xanthine, one mild stimulant and bronchodilator, was used to treat asthma especially, which can produce  $\text{H}_2\text{O}_2$  in the presence of xanthine oxidase. Based on the peroxidase-like activity of the Ir NPs,  $\text{H}_2\text{O}_2$  and xanthine were analyzed using

\* Corresponding author.

E-mail address: [qsong@jiangnan.edu.cn](mailto:qsong@jiangnan.edu.cn) (Q. Song).

TMB as colorimetric substrate. Furthermore, the Michaelis-Menten behavior of the Ir NPs was investigated according to the steady-state kinetics analysis, and the catalytic mechanism of the Ir NPs was proposed.

## 2. Materials and methods

### 2.1. Materials

Iridium (III) chloride trihydrate ( $\text{IrCl}_3 \cdot 3\text{H}_2\text{O}$ ), 3,3',5,5'-tetramethylbenzidine (TMB), artemisinin and xanthine were purchased from Aladdin Industrial Co., Ltd. (Shanghai, China). Xanthine oxidase was purchased from Sigma-Aldrich (Shanghai, China). Horseradish peroxidase (HRP) was obtained from Sangon Biotech Co., Ltd. (Shanghai, China). Spin trap 5,5-dimethyl-1-pyrroline N-oxide (DMPO, 97%) was purchased from Energy Chemical Co., Ltd. (Shanghai, China). Tannic acid, sodium borohydride ( $\text{NaBH}_4$ ), hydrogen peroxide ( $\text{H}_2\text{O}_2$ ), ethanol ( $\text{C}_2\text{H}_5\text{O}$ ), acetic acid (HAc), sodium acetate anhydrous (NaAc), o-Phenylenediamine (OPD) and pyrogallol were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). All reagents were of analytical grade and were used without further purification. Milli-Q ultrapure water (18.2 M $\Omega$ ) was used throughout all experiments.

### 2.2. Instruments

Absorption spectra and peroxidase kinetics assay were carried out on a UV–vis spectrophotometer with a 1 cm quartz cuvette (UV-2700, Japan Shimadzu Co., Ltd). High resolution transmission electron microscopy (HRTEM, JEM-2100, Japan electron optics laboratory co., Ltd) was performed at 200 kV to visualize the prepared Ir NPs, and Energy dispersive X-ray (EDX) analysis of the Ir NPs were carried out with JEM-2100 (HR) microscope. The samples for TEM measurement were prepared by slow evaporation of the Ir NPs solution on a carbon film supported by 300-mesh Cu grids. The hydrodynamic diameter of the Ir NPs was measured by dynamic light scattering (DLS) and the Zeta potential was determined by using the Zeta potential analyzer (Brookhaven instruments Corporation). The oxidation state of iridium and the interaction with tannic acid were analyzed by X-ray photoelectron spectroscopy (XPS) using a PHI Quantum 2000 Scanning ESCA Microprobe. Phase identification of the Ir NPs were conducted with X-ray diffraction (XRD, D8, Bruker AXS Co., Ltd) using Cu-K $\alpha$  radiation source ( $\lambda = 1.54051 \text{ \AA}$ ) over the  $2\theta$  range of 3–90°. All Electron Spin Resonance (ESR) measurements were carried out using a Bruker EMXplus-10/12 spectrometer at ambient temperature. ESR parameter settings were as follows: modulation amplitude 1 G, scan range 100 G, Center Field 3500 G, StaticFieldMon 3450 G, modulation frequency 100 kHz and microwave power 20 mW for the determination of spin adducts using spin traps DMPO.

### 2.3. Preparation of the Ir NPs

The Ir NPs were prepared by a simple hydrothermal process using Iridium (III) chloride ( $\text{IrCl}_3$ ) as precursor and tannic acid as stabilizer. In brief,  $\text{IrCl}_3$  solution (5.00 mL, 2.0 mM) was added to tannic acid solution (5.00 mL, 2.0 mM) under vigorous stirring and the color of the mixed solution immediately changed into faint yellow, indicating the formation of Ir-tannic acid complex. After boiling the mixture solution for 30 min, freshly prepared  $\text{NaBH}_4$  solution (1 mL, 0.5 M) was added and the mixture was stirred for 3 h. The color of the solution changed from faint yellow to deep brown gradually, indicating the formation of the colloid Ir NPs. Finally, the obtained Ir NPs was stored at ambient temperature for use.

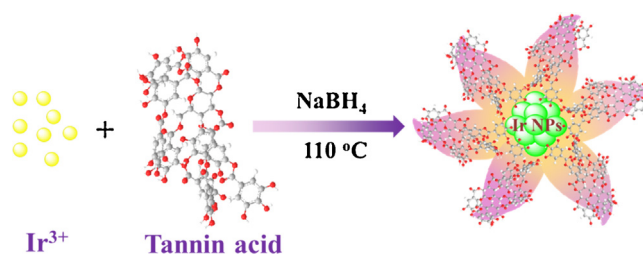


Fig. 1. Schematic of the Ir NPs preparation with one-step production.

### 2.4. Kinetic analysis

The Michaelis-Menten behavior of the Ir NPs was investigated by monitoring the absorbance of TMB at 652 nm with UV–vis spectrophotometer. The kinetic measurements were conducted in 5 mL NaAc buffer solution (20 mM, pH 4, 35 °C) containing 38.44 ng/mL Ir NPs, with various concentrations of TMB or  $\text{H}_2\text{O}_2$ . The Michaelis-Menten constant was calculated using Lineweaver-Burk plots,  $1/v = (K_m/V_{\max})(1/[S]) + 1/V_{\max}$ , where  $v$  represents the initial velocity,  $V_{\max}$  stands for the maximal reaction velocity,  $[S]$  is the concentration of substrate. The  $K_m$  is the Michaelis constant, which is an indication for the affinity of the Ir NPs mimics with the substrate.

### 2.5. Catalytic mechanism study

#### 2.5.1. Electron spin resonance

The catalytic mechanism of the Ir NPs was evaluated in the  $\text{H}_2\text{O}_2$ /UV/DMPO system in the presence and absence of Ir NPs. Samples containing 20 mM DMPO, 10 mM  $\text{H}_2\text{O}_2$ , and different concentrations of the Ir NPs (0, 38.44, 76.88, 153.76 ng/mL) were prepared in NaAc buffer (20 mM, pH 4). Each sample was UV-irradiated at 365 nm for 10 min, and then transferred to a quartz capillary tube and placed in the ESR cavity for spectra recording.

#### 2.5.2. Catalytic performance comparison of the Ir NPs against benzoyl peroxide, $\text{H}_2\text{O}_2$ and artemisinin

Samples containing 100  $\mu\text{M}$  TMB, 38.44 ng/mL Ir NPs, and different peroxides (benzoyl peroxide,  $\text{H}_2\text{O}_2$  and artemisinin) were prepared in NaAc buffer (20 mM, pH 4), which were then incubated for various time in ambient temperature. The color changes and the absorbance spectra were recorded for subsequent analysis.

### 2.6. Determination of $\text{H}_2\text{O}_2$ and xanthine

For the determination of  $\text{H}_2\text{O}_2$ , an aliquot of 5 mL NaAc buffer solution (20 mM, pH 4) containing 250  $\mu\text{M}$  TMB, 38.44 ng/mL Ir NPs and different concentrations of  $\text{H}_2\text{O}_2$  was incubated at 35 °C for 10 min for UV–vis measurement.

For the determination of xanthine, 1.0 mL reaction solution containing  $5 \times 10^{-4}$  U xanthine oxidase and different concentrations of xanthine was incubated for 2 h under ambient temperature. Then, 10  $\mu\text{L}$  19.22  $\mu\text{g/mL}$  Ir NPs, 250  $\mu\text{L}$  5 mM TMB and 1.0 mL 100 mM NaAc buffer were added. The reaction solutions were diluted to 5.0 mL with water and incubated at 35 °C for 10 min for UV–vis measurement.

## 3. Results and discussion

### 3.1. Preparation and characterization of the Ir NPs

The Ir NPs were prepared with one-pot reaction in aqueous solution using tannic acid as stabilizer (Fig. 1). Two adjacent hydroxyl groups in tannic acid could complex with  $\text{Ir}^{3+}$  to form five-member

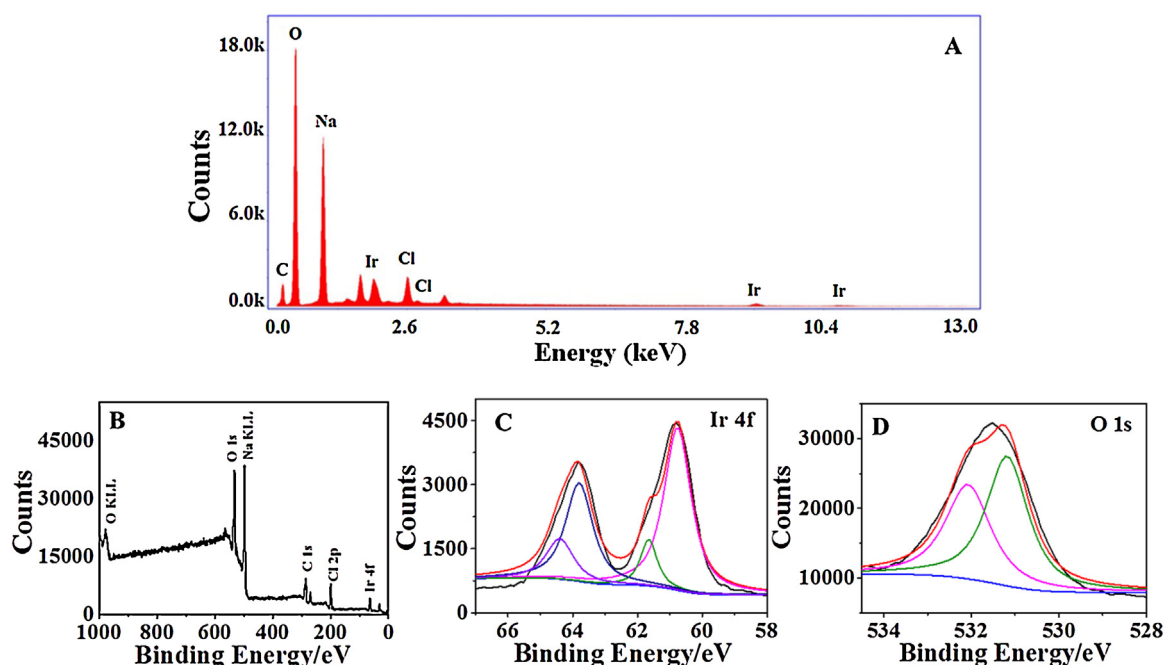


Fig. 2. (A) EDX spectrum of the Ir NPs, (B) XPS spectrum of the Ir NPs, (C) deconvoluted Ir 4f and (D) deconvoluted O 1s regions of the spectrum.

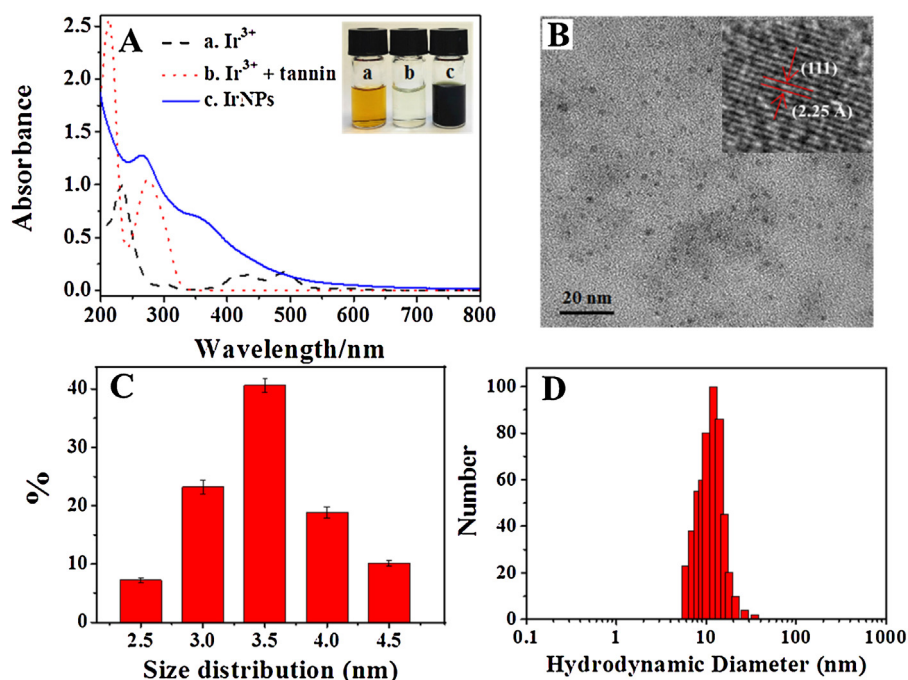


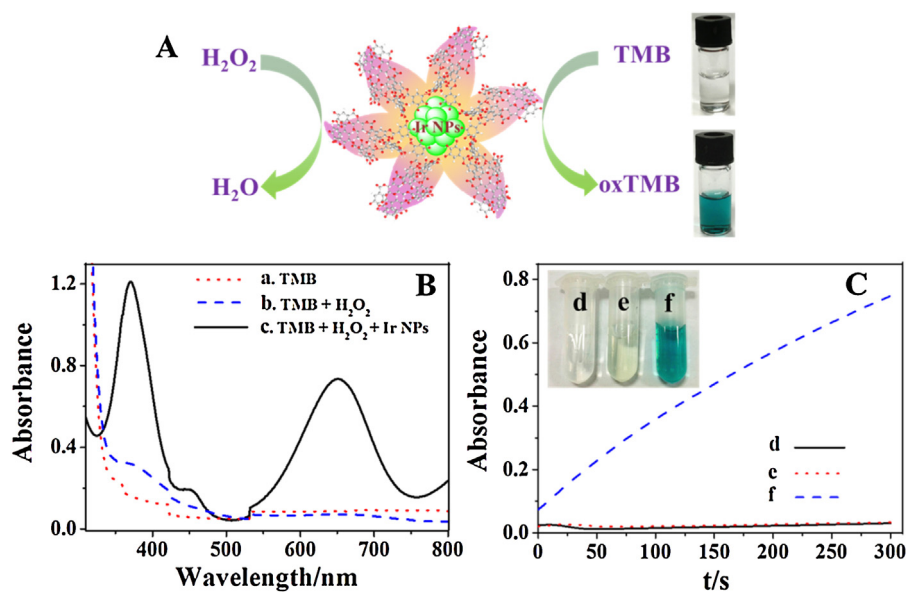
Fig. 3. (A) UV-vis absorbance spectra of  $\text{Ir}^{3+}$ , tannin acid and the Ir NPs, (B) HRTEM images of the Ir NPs, the inset of figure B shows the lattice spacing, (C) Size distribution analysis of the Ir NPs from 89 random nanoparticles, (D) Hydrodynamic diameter of the Ir NPs measured by DLS.

ring in the beginning of the reaction. After the addition of  $\text{NaBH}_4$ , the chelated  $\text{Ir}^{3+}$  was reduced to form Ir NPs [17,18]. The number of the nanoparticles increased gradually with the increasing of  $\text{NaBH}_4$  concentration from 0.1 M to 0.5 M, and aggregation could be observed with further increased the concentration of  $\text{NaBH}_4$  (Fig. S1).

The EDX spectrum of the Ir NPs verified the presence of iridium (Fig. 2A), and the oxidation state of Ir and the interaction with tannin acid were confirmed through XPS analysis (Fig. 2B). As shown in Fig. 2C, two peaks at 60.8 and 63.8 eV belong to the binding energies of  $4f_{7/2}$  and  $4f_{5/2}$  of Ir(0), respectively, and two peaks at 61.7

and 64.5 eV of  $\text{Ir}^{\delta+}\text{-O}$  [15]. The two peaks of O 1s spectra, 531.2 and 532.1 eV correspond to  $\text{C=O}$  and  $\text{Ir}^{\delta+}\text{-O}$ , respectively (Fig. 2D). These results suggest that the iridium cations on the surface of the Ir NPs bind with the phenolic hydroxyls of tannin acid.

As shown in Fig. 3A, the pure  $\text{Ir}^{3+}$  solution showed four obvious absorbance peaks at 233, 305, 421 and 486 nm, respectively. After mixed with tannin acid solution, these four absorbance peaks disappeared, indicating that  $\text{Ir}^{3+}$  complexed with the hydroxyl of tannin acid [17]. With the reduction by  $\text{NaBH}_4$ , two new absorbance peaks appeared at 261, 363 nm and other bands disappeared, suggesting the formation of the Ir NPs. From the photographs inserted



**Fig. 4.** (A) Schematic illustration of color reaction of TMB oxidized by H<sub>2</sub>O<sub>2</sub> under the catalytic action of the Ir NPs; (B) UV-vis absorbance spectra of (a) TMB, (b) H<sub>2</sub>O<sub>2</sub> + TMB, (c) H<sub>2</sub>O<sub>2</sub> + TMB + Ir NPs; (C) Time-dependent absorbance of TMB at 652 nm in different reaction systems: (d) H<sub>2</sub>O<sub>2</sub> + TMB, (e) TMB + Ir NPs, (f) H<sub>2</sub>O<sub>2</sub> + TMB + Ir NPs. Insert is the photos of different reaction systems. Reaction conditions: 250  $\mu$ M TMB, 2 mM H<sub>2</sub>O<sub>2</sub>, 38.44 ng/mL Ir NPs, 20 mM NaAc buffer (pH 4), 35  $^{\circ}$ C.

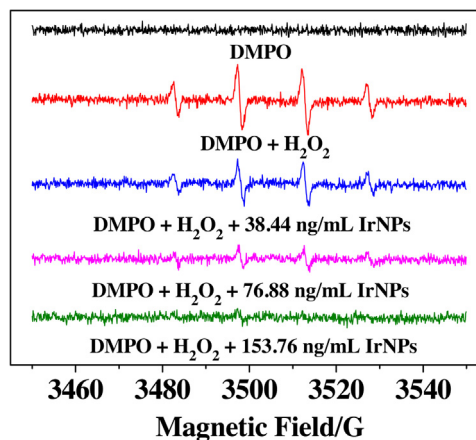
in Fig. 3A, the color changes of solutions also indicated the successful preparation of the Ir NPs.

Fig. 3B displayed the HRTEM image of the Ir NPs and the inset revealed that the crystal lattice fringe was 2.25 Å, which corresponds to the lattice constant of (111) face of the metallic Ir. The well monodispersed and uniform Ir NPs showed an average diameter of  $3.5 \pm 0.5$  nm based on the statistic result of 89 particles, while the hydrodynamic diameter was about 12 nm measured by DLS (Fig. 3C, D). These results suggest that the tannin acid can provide the steric hindrance and protect the Ir NPs from the aggregation. The large surface area of small particles and the excellent dispersity endowed the Ir NPs with superior catalytic activity. Selected area electron diffraction and XRD patterns indicated the amorphous structure of the Ir NPs (Fig. S2). As shown in Fig. S3, the Zeta potential of the Ir NPs was determined to be  $-54.8$  mV, indicating that the surface of the Ir NPs was negatively charged. The absolute value was much greater than 20 mV, thus Ir NPs can maintain excellent stability due to their strong electrostatic repulsion effects [19].

### 3.2. Kinetic assay of the Ir NPs catalytic reaction

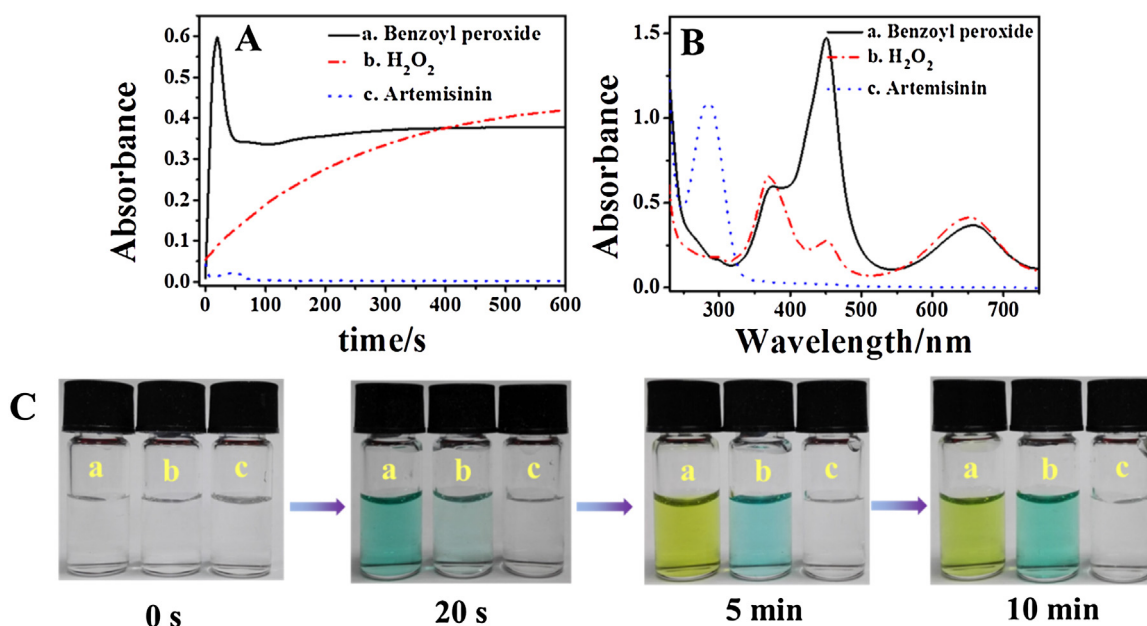
In order to investigate the peroxidase-like activity of the Ir NPs, the catalytic oxidation of peroxidase substrates TMB, OPD and pyrogallol were studied in the presence of H<sub>2</sub>O<sub>2</sub>. In pH 4 NaAc buffer, three substrates were oxidized by H<sub>2</sub>O<sub>2</sub> under the catalytic action of the Ir NPs and produced the typical color reaction (Fig. S4). H<sub>2</sub>O<sub>2</sub> or Ir NPs alone did not oxidized substrates to generate corresponding color reactions. Taking the case of TMB, the Ir NPs can catalyze the oxidation of TMB by H<sub>2</sub>O<sub>2</sub> to produce blue product with characteristic absorbance peak at 652 nm (Fig. 4A, B). As shown in Fig. 4C, the time-dependent absorbance of H<sub>2</sub>O<sub>2</sub>-TMB-Ir NPs system at 652 nm was much higher than that observed in the TMB-H<sub>2</sub>O<sub>2</sub> and TMB-Ir NPs systems, indicating the effective catalytic action of Ir NPs to H<sub>2</sub>O<sub>2</sub>.

The catalytic activity of the Ir NPs was dependent on pH and temperature (Figs. S5a and b). Hydrogen ion is one of the reactants, and thus the increase of acidity can speed up the reaction rate and facilitate the oxidation of TMB [20]. The proton concentration also influences the chemical reactivity of TMB, because the ionization of amino groups in TMB is pH-dependent. Alkaline environment



**Fig. 5.** Effect of the Ir NPs on the formation of  $\cdot$ OH in H<sub>2</sub>O<sub>2</sub>/UV system. Samples containing 20 mM DMPO, 10 mM H<sub>2</sub>O<sub>2</sub>, 20 mM NaAc buffer (pH 4) and different Ir NPs concentration were exposed under 365 nm UV light for 10 min.

could inhibit the combination and dissociation rates between the substrate and the Ir NPs, hence decreasing the catalytic activity of the Ir NPs [21]. In case of temperature influence, the TMB may decompose when the temperature was over 55  $^{\circ}$ C, whereas for temperature below 35  $^{\circ}$ C, a decrease in the catalytic activity of the Ir NPs was also observed, presumably due to the decrease in the collision rate between the TMB with the Ir NPs. These results suggested that the Ir NPs exerted the optimal peroxidase-like activity at pH 4.0 under 35  $^{\circ}$ C. The relative activity of the Ir NPs reached the maximum in the presence of 5 mM H<sub>2</sub>O<sub>2</sub>, further increase in H<sub>2</sub>O<sub>2</sub> concentration can inhibit their peroxidase-like activity (Fig. S5c). These behaviors were similar to that observed with HRP and further confirmed the peroxidase-like activity of the Ir NPs [22]. In addition, the relative activity of the Ir NPs and HRP incubated at different conditions were compared (Fig. S6a and b). Results showed the relative activity of the Ir NPs was substantially higher than that of HRP. Furthermore, the Ir NPs retained over 90% of their initial catalytic activity after being stored for 3 months in ambient temperature (Fig. S6c), showing the superior stability of the Ir NPs against change of environment conditions.



**Fig. 6.** (A) Time-dependent absorbance of TMB at 652 nm, (B) UV-vis absorbance spectra and (C) Photos of time-dependent color changes of different reaction systems (a) TMB + benzoyl peroxide, (b) TMB + H<sub>2</sub>O<sub>2</sub> + Ir NPs, (c) TMB + artemisinin + Ir NPs. Reaction conditions: 100  $\mu$ M TMB, 0.05 mM benzoyl peroxide, 0.5 mM H<sub>2</sub>O<sub>2</sub>, 5 mM artemisinin, 38.44 ng/mL Ir NPs, 20 mM NaAc buffer (pH 4), 35  $^{\circ}$ C.

**Table 1**

Comparison of the apparent Michaelis-Menten constants ( $K_m$ ) and maximum reaction rates ( $V_{max}$ ) between the Ir NPs and HRP.

| Catalyst | Substrates                    | $K_m$ /mM | $V_{max}/10^{-7} \text{ M s}^{-1}$ |
|----------|-------------------------------|-----------|------------------------------------|
| Ir NPs   | H <sub>2</sub> O <sub>2</sub> | 18.02     | 0.81                               |
| Ir NPs   | TMB                           | 0.03      | 0.17                               |
| HRP      | H <sub>2</sub> O <sub>2</sub> | 0.38      | 0.57                               |
| HRP      | TMB                           | 0.53      | 1.45                               |

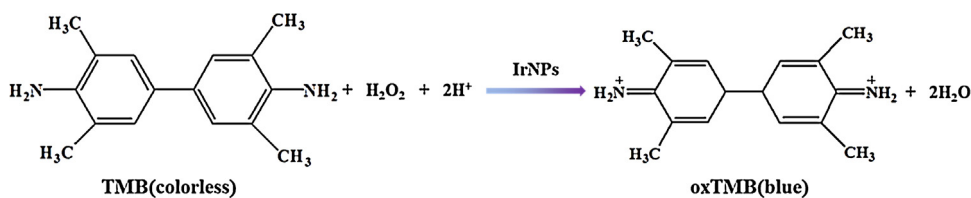
The Michaelis-Menten behavior of the Ir NPs was evaluated by steady-state kinetics analysis. A series of initial reaction rates were obtained by recording the absorbance of TMB at 652 nm when varying one substrate concentration and keeping the other substrate concentration constant. The steady-state kinetic parameters  $K_m$  and  $V_{max}$  were acquired by fitting these initial reaction rates to Lineweaver-Burk plot  $1/v = (K_m/V_{max})(1/[S]) + 1/V_{max}$ . As shown in Fig. S7a–d, the oxidation reaction catalyzed by the Ir NPs followed the typical Michaelis-Menten behavior. As shown in Fig. S7e and f, the parallel double reciprocal plots suggested a ping-pong mechanism and indicated that the Ir NPs bound and reacted with first substrate and then released the first product before reacted with the other substrates [23,24]. As shown in Table 1, the  $K_m$  of HRP for TMB oxidation was 17.67 times higher than that of the Ir NPs, indicating apparently higher affinity of the Ir NPs to TMB than HRP. The  $K_m$  value of the Ir NPs for H<sub>2</sub>O<sub>2</sub> reduction was 47 times higher than that of HRP, suggesting that the Ir NPs can endure higher H<sub>2</sub>O<sub>2</sub> concentration than HRP.

### 3.3. Catalytic mechanism

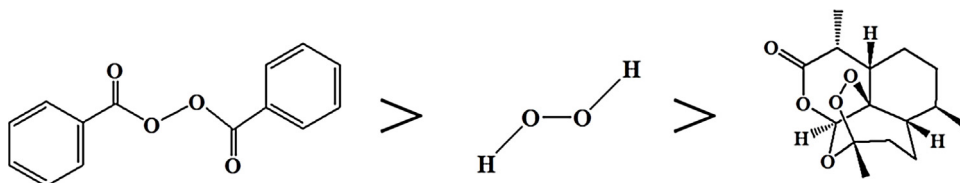
In order to investigate the catalytic mechanism of the Ir NPs peroxidase mimics, two approaches were designed. First, H<sub>2</sub>O<sub>2</sub> can generate hydroxyl radical ( $\cdot\text{OH}$ ) under UV irradiation and spin trap DMPO can trap the  $\cdot\text{OH}$  to form the stable DMPO/ $\cdot\text{OH}$  spin adduct, which yields the typical four lines ESR spectrum with relative intensities 1:2:2:1. ESR signal intensity will attenuate with the addition of the Ir NPs in a concentration-dependent manner (Fig. 5). The experiment results indicated that the peroxidase-like activity of Ir NPs does not originate from the production of  $\cdot\text{OH}$ .

Second, the catalytic performance of the Ir NPs for peroxide substrates with different electron withdrawing ability was investigated. The time-dependent absorbance obtained from TMB-benzoyl peroxide, TMB-H<sub>2</sub>O<sub>2</sub>-Ir NPs, and TMB-artemisinin-Ir NPs systems were compared and the results indicated that the catalytic reducing rate of the Ir NPs for H<sub>2</sub>O<sub>2</sub> was faster than artemisinin and slower than benzoyl peroxide (Fig. 6A). Even in the absence of the Ir NPs, the TMB can be completely oxidized by benzoyl peroxide in 20 s. For artemisinin, this reaction could not occur in the presence of the Ir NPs. The absorbance peak of TMB-benzoyl peroxide system at 652 nm reached the maximum value in 20 s and then gradually decreased, meanwhile the absorbance peak at 450 nm gradually increased (Fig. 6B(a)). The results were in accordance with the color changes presented in Fig. 6C, suggesting that TMB was firstly oxidized by benzoyl peroxide to generate blue production and further was oxidized to form yellow production. As shown in Fig. 6B(c), no obvious absorbance peak at 652 and 450 nm appeared in TMB-artemisinin-Ir NPs system, implied that artemisinin can not oxidize TMB even in the presence of the Ir NPs.

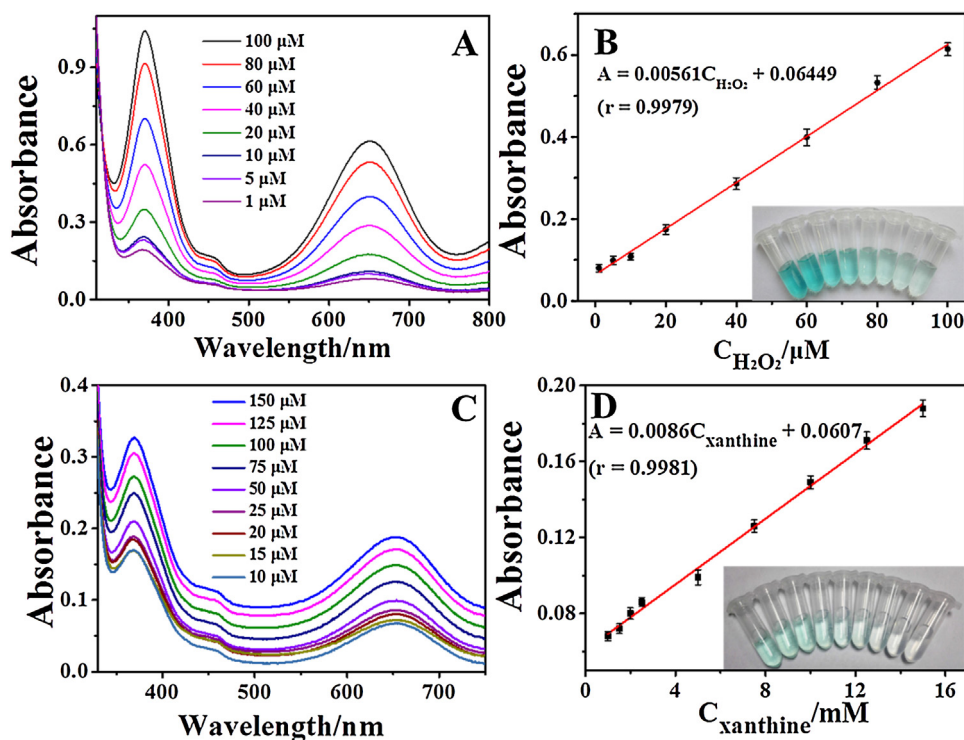
Seen from the molecule structures of three peroxides, the electron withdrawing ability of H<sub>2</sub>O<sub>2</sub> was higher than artemisinin and lower than benzoyl peroxide (Schemes 1 and 2). Thus, we speculated that the Ir NPs could facilitate the electron transfer between TMB and peroxides. TMB was absorbed on the surface of the Ir NPs and donated lone-pair electrons from the amino groups to the Ir NPs, resulting in the increase of electron density of the Ir NPs [25,26]. Then the electrons transferred from the Ir NPs to peroxides and facilitated the oxidation of TMB [25,27]. Thus the Ir NPs was much easier to catalyze the oxidation of TMB by H<sub>2</sub>O<sub>2</sub> than



**Scheme 1.** The corresponding reaction equation of TMB oxidized by  $\text{H}_2\text{O}_2$  in presence of the Ir NPs.



**Scheme 2.** Comparison of electron withdrawing ability of  $\text{H}_2\text{O}_2$ , artemisinin and benzoyl peroxide.



**Fig. 7.** UV-vis absorbance spectra of TMB oxidized by different concentration of  $\text{H}_2\text{O}_2$  (A) and xanthine (C) under the catalytic action of the Ir NPs. The linear calibration plots for  $\text{H}_2\text{O}_2$  (B) and xanthine (D) determination using the Ir NPs as peroxidase mimics. Inset: the photographs of TMB oxidized by different concentration of  $\text{H}_2\text{O}_2$  (from left to right: 100–1  $\mu\text{M}$ ) (B) and xanthine (from left to right: 150–10  $\mu\text{M}$ ) (D) under visible light.

artemisinin, but more difficult than benzoyl peroxide, due to the difference in the ability of accepting electrons.

#### 3.4. Determination of $\text{H}_2\text{O}_2$ and xanthine

The color changes of TMB in presence of the Ir NPs were explored for the determination of  $\text{H}_2\text{O}_2$  based on the relationship between the absorbance of TMB at 652 nm and the  $\text{H}_2\text{O}_2$  concentrations (Fig. 7A). The absorbance value at 652 nm increased proportionately with the increasing of  $\text{H}_2\text{O}_2$  concentration from 1 to 100  $\mu\text{M}$ . The calibration plot demonstrated a good linear relation (Fig. 7B). The regression equation was  $A = 0.06649 + 0.00561C_{\text{H}_2\text{O}_2}$ , with the correlation coefficient ( $r$ ) of 0.9979 and the limit of detection (LOD) of 0.53  $\mu\text{M}$  based on the  $3\sigma$  criterion. Xanthine can be catalyzed by xanthine oxidase to produce  $\text{H}_2\text{O}_2$ , hence the Ir NPs were also

applied in the colorimetric and quantitative determination of xanthine (Fig. 7C). As depicted in Fig. 7D, the absorbance of TMB at 652 nm was proportional to the xanthine concentration in the range from 10 to 150  $\mu\text{M}$  with the LOD of 5.2  $\mu\text{M}$ . The linear regression equation was  $A = 0.0607 + 0.0086C_{\text{xanthine}}$  with the  $r$  of 0.998. The color changes from colorless to blue can be observed by the naked eyes (inset of Fig. 7B and D). These results indicated that the Ir NPs can be used as colorimetric sensor for the determination of  $\text{H}_2\text{O}_2$  and xanthine.

As shown in Table 2, compared with other sensors for  $\text{H}_2\text{O}_2$  assay, the Ir NPs sensor showed high sensitivity and large calibration range. Especially our method also showed the other advantages, including simple preparation, easy operation and fast response time. As for the determination of xanthine, the sensitivity was higher than electrochemical methods [33,35] and lower than

**Table 2**Comparison of various sensors for the determination of H<sub>2</sub>O<sub>2</sub> and xanthine.

| Material                                                             | Analysts                      | Method          | Linear range/ $\mu$ M           | LOD/ $\mu$ M           | Ref.        |
|----------------------------------------------------------------------|-------------------------------|-----------------|---------------------------------|------------------------|-------------|
| N-GQDs <sup>a</sup>                                                  | H <sub>2</sub> O <sub>2</sub> | Colorimetric    | 20–1170                         | 5.3                    | [28]        |
| Nanoceria                                                            |                               | Colorimetric    | 4–40                            | 2.5                    | [29]        |
| Au@Pt NRS <sup>b</sup>                                               |                               | Colorimetric    | 45–3000                         | 45                     | [30]        |
| PtPd nanodendrites                                                   |                               | Colorimetric    | 0.5–150                         | 0.1                    | [31]        |
| Ce <sub>3</sub> O <sub>4</sub> NPs                                   |                               | Electrochemical | 50–2.5 $\times$ 10 <sup>4</sup> | 10                     | [32]        |
| ZnO NPs                                                              |                               | Fluorescence    | 50–10 <sup>5</sup>              | 50                     | [33]        |
| Ir NPs                                                               |                               | Colorimetric    | 1–100                           | 0.53                   | This method |
| XO <sup>c</sup> /Co <sub>3</sub> O <sub>4</sub> /MWCNTs <sup>d</sup> | Xanthine                      | Electrochemical | Anode: 0.2–16 Cathode: 20–360   | Anode: 0.2 Cathode: 20 | [34]        |
| XO/PtPd/graphite electrode                                           |                               | Electrochemical | 0–40                            | 4.5                    | [35]        |
| PVF <sup>e</sup> +XO <sup>-f</sup> /Pt electrode                     |                               | Electrochemical | 430–2840                        | 130                    | [36]        |
| Au NPs                                                               |                               | Colorimetric    | 0.125–6                         | 0.023                  | [37]        |
| Ir NPs                                                               |                               | Colorimetric    | 10–150                          | 5.2                    | This method |

<sup>a</sup> N-GQDs: nitrogen-doped graphene quantum dots.<sup>b</sup> Au@Pt NRS: Au nanorods @ Pt nanodots core/shell nanostructures.<sup>c</sup> XO: xanthine oxidase.<sup>d</sup> MWCNTs: multiwalled carbon nanotube.<sup>e</sup> PVF<sup>+</sup>: polyvinylferrocenium cation.<sup>f</sup> XO<sup>-</sup>: xanthine oxidase anion.

the colorimetric method based on Au NPs sensor [36], presumably due to the low yield of H<sub>2</sub>O<sub>2</sub> production from the oxidation reaction of xanthine.

#### 4. Conclusions

In conclusion, the Ir NPs peroxidase mimics were prepared by one-pot synthesis using tannin acid as stabilizer in aqueous solution. The Ir NPs exhibited the high peroxidase-like activity and can catalytically oxidize TMB, OPD and pyrogallol using H<sub>2</sub>O<sub>2</sub> to produce color reaction. The catalytic mechanism was based on the electron density change and mobility in the Ir NPs. The catalytic behavior was in accordance with Michaelis-Menten kinetics and followed a ping-pong mechanism. The steady-state kinetic analysis indicated the Ir NPs showed much higher affinity to TMB and can endure higher H<sub>2</sub>O<sub>2</sub> concentration than HRP. The Ir NPs peroxidase mimics possessed higher stability against pH and temperature change than HRP, and the peroxidase-like activity of the Ir NPs was over 90% after storage for 3 months. Due to the excellent water solubility, stability and non-toxicity, the Ir NPs can be potentially used as peroxidase mimics for biomedical field.

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#### Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.snb.2016.11.145>.

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## Biographies

**Malin Cui** (female) is studying for Ph.D. in School of Chemical and Material Engineering at Jiangnan University. She received her M.S. in Department of Analytical Chemistry at Minnan Normal University in 2013. Currently her research involves the synthesis and applications of noble metal iridium nanomaterials.

**Jingdan Zhou** (female) is studying for M.S. in School of Chemical and Material Engineering at Jiangnan University. She received the B.S. in Department of Chemical and Material Engineering at Hangzhou Normal University in 2014.

**Yuan Zhao** (female) is an associate professor of School of Chemical and Material Engineering at Jiangnan University, Wuxi, China. She completed her Ph.D. from School of Food Science and Technology at Jiangnan University in 2013. Her research interests include the synthesis and analytical applications in sensors and catalysts of Ru, Au, Ag nanomaterials.

**Qijun Song** (male) is a professor of School of Chemical and Material Engineering at Jiangnan University, Wuxi, China. He got his Ph.D. from Chemistry department, the University of Hull in 2003. His current researches are focused on the development of novel analytical methods and their applications in environment, food and forensic analysis.